

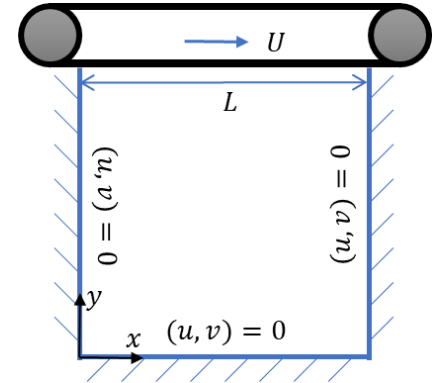
Assignment 4: Computational Fluid Dynamics (ME543), Aug-Nov 2024

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- Provide: (i) the grid detail, (ii) the discretized equations detail, (iii) the boundary condition implementation detail, (iv) the required output (plots/any other such means), (v) a well-documented code.
- Items (i), (ii), and (iii) above should be written out/typed on a separate sheet and attached before items (iv). Soft copy of document containing items (i) to (iv) and code need to be submitted.
- Solve the following problems using finite difference method. If not provided, you can decide your own grid spacing and determine roughly the *coarsest* grid for *sufficient* accuracy, say 10^{-6} .
- You are advised to choose a non-uniform mesh.
- Solve the following 2-D incompressible flow problems using stream function-vorticity formulation:

1. Solve the 2-D lid-driven cavity problem with the specified boundary conditions on a 128×128 grid size as shown in the figure. Compare the results with Ghia et. al [1] for $Re = \rho UL/\mu = 100$ and 400 in terms of

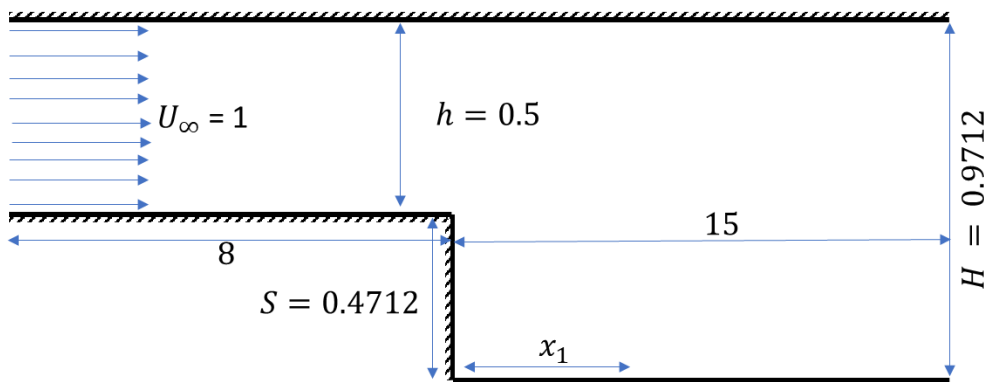
- a) streamlines,
- b) velocity vectors,
- c) vorticity contours
- d) u velocity along vertical centerline and,
- e) v velocity along horizontal centerline.



2. Solve the 2-D flow over a backward facing step for the geometry shown in the figure. Take $Re = \rho U_\infty D/\mu = 100$, where $D = 2h$ is the hydraulic diameter. Find the recirculation length (x_1) and plot

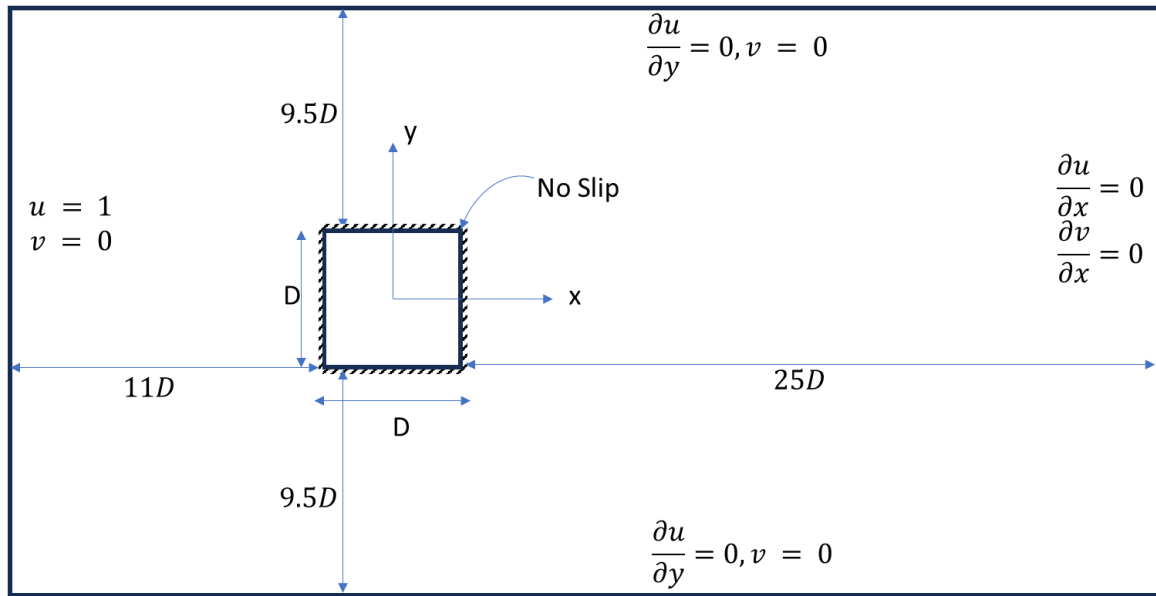
- a) velocity vectors,
- b) stream function and vorticity contours
- c) u velocity along vertical line at separation length. (Refer to [2] Figure 15 for velocity at separation length)

Refer to [2] and a software called “webplot digitizer” for extracting the x -component of velocity from paper and comparing it with your results. Recirculation length is the distance from step where the x -component of velocity changes sign near bottom wall [2].



3. (50% additional marks) Solve the flow around a square cylinder at $Re = \rho U_\infty D/\mu = 100$ for the geometry shown in the figure.

- a) Please show the mesh chosen for the problem.
- b) Plot vorticity contours at $t = 300$.
- c) Plot variation of Coefficient of Lift and drag with time and find CD_{mean} and rms lift force coefficient (Refer Table 3 in [3]).



Reference:

1. U. Ghia, K.N. Ghia, and C. T. Shin, "High-Resolutions for Incompressible Flow Using the Navier-Stokes Equations", Journal of Computational Physics, vol. 48, pp. 387-411, 1982.
2. B. F. Armaly, F. Durst, Jose C.f. Pereira, B. Schonung, "Experimental and Theoretical Investigation of Backward-Facing Step Flow", Journal of Fluid Mechanics-127:473 – 496, 1983.
3. A. Sohankar, C. Norberg, and L. Davidson, "Numerical simulation of unsteady flow around a square two-dimensional cylinder," Twelfth Australasian Fluid Mechanics Conference, 1995.

Coding guidelines/tips

- Refer assignment 1.